Correcting Sight

Exam code: 0625 & 0972

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Correcting sight

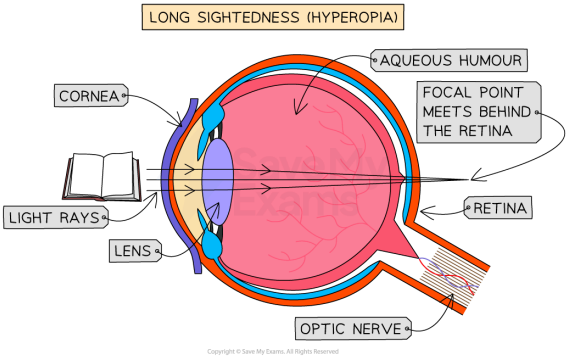
Extended tier only

- Converging and diverging lenses are commonly used in glasses and contact lenses to correct defects of sight
  - Converging lenses can be used to correct long-sighted vision
  - Diverging lenses can be used to correct short-sighted vision

Use of lenses to correct long-sightedness

- Long-sighted people have eyes that are **less curved** than normal or the eyeball is **too short**
  - This means they cannot see things that are close and can only clearly see things that are far away

Ray diagram to show long-sightedness

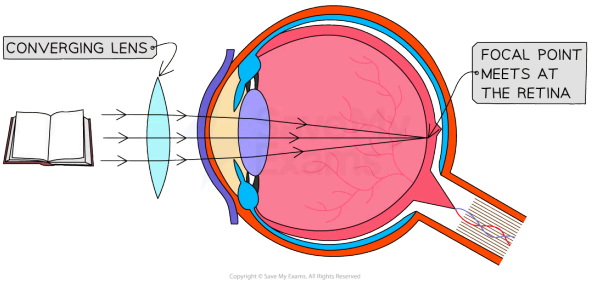


An eye that is long-sighted has a narrower lens with a smaller focussing power so the light rays meet and form an image behind the retina and not on it

- The eye refracts the light rays and they are brought to a focus **beyond** the retina
  - In other words, the focus point is behind the retina at the back of the eye
- This can be corrected by using a convex or **converging lens**

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

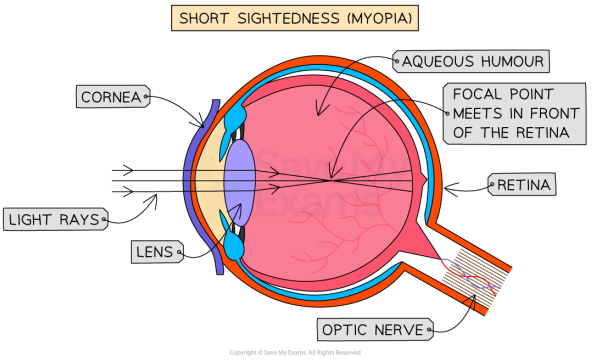


The converging lens causes the rays to converge before they reach the eye, so the image is formed on the retina and not behind it

Use of lenses to correct short-sightedness

- People who are **short-sighted** have eyes that are **more curved** than normal or have an eyeball that is **too long**
  - This means they cannot see things that are far away, and only see things that are close to them

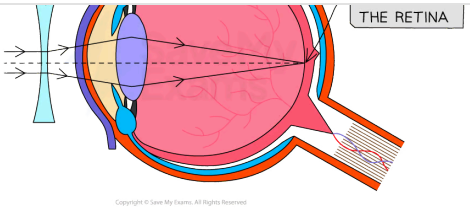
Ray diagram of short-sightedness



An eye that is short-sighted has a wider lens with a larger focussing power so the light rays meet and form an image in front of the retina and not on it

- This is because the eye refracts the light and brings it to a focus **before** it reaches the retina
  - In other words, the focus point is in front of the retina at the back of the eye
- This can be corrected by using a concave or a **diverging lens**

The effect of a diverging lens on a short-sighted eye



The diverging lens causes the rays to diverge before they reach the eye, so the image is formed on the retina and not in front of it

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